

NTC_CC105 – DATA STRUCTURE W/LAB

WEEK 5 LABORATORY INSTRUCTION

Laboratory Activity 5

Instructor	Prof. Darwin C. Vargas, MIT
Program	BSIT
Academic Year	2026–2027
Programming Language	C++

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4. Date	Oct 5, 2026
5. Laboratory Activity No.	Laboratory Activity 5
6. Laboratory Title	Linear Architecture Synthesis & Peer Code Audit

PART A: Peer Code Audit Checklist Table

Audit Category	Specific Issue Located in Legacy Code	Line Number(s)	Severity (Low / Med / High / Critical)	Impact / Risk
Global State	Mutable variables (dataArray , currentCount , maxCapacity).	6 - 8	High	Prevents multiple instances; thread-unsafe.
Memory Leak	Reallocates temp without calling delete[] dataArray .	17 -23	Critical	Causes RAM exhaustion on every resize.
Memory Leak	Subsystem exits without freeing dataArray .	103 - 106	High	Leaves dynamic memory unreleased at termination.
Out of Bounds	Print loop condition i <= currentCount reads past array.	63 - 66	High	Reads garbage data; risks access violations.
Out of Bounds	remove_item_at() lacks upper bound check (idx >= currentCount).	33 - 36	Critical	Corrupts memory or crashes on invalid index removal.
Inefficient Shift	Redundant inner loop inside element-shifting routine.	39 - 44	Medium	Slows down deletion performance unnecessarily.
Naming Standards	Inconsistent mixed naming (add_item vs findItem).	10 / 31 / 49 / 61	Low	Breaks code readability standards.
Hardcoded Sizing	Fixed 3x3 array bounds in processMatrix() .	74 - 78	Medium	Restricts matrix reusability to a single size.